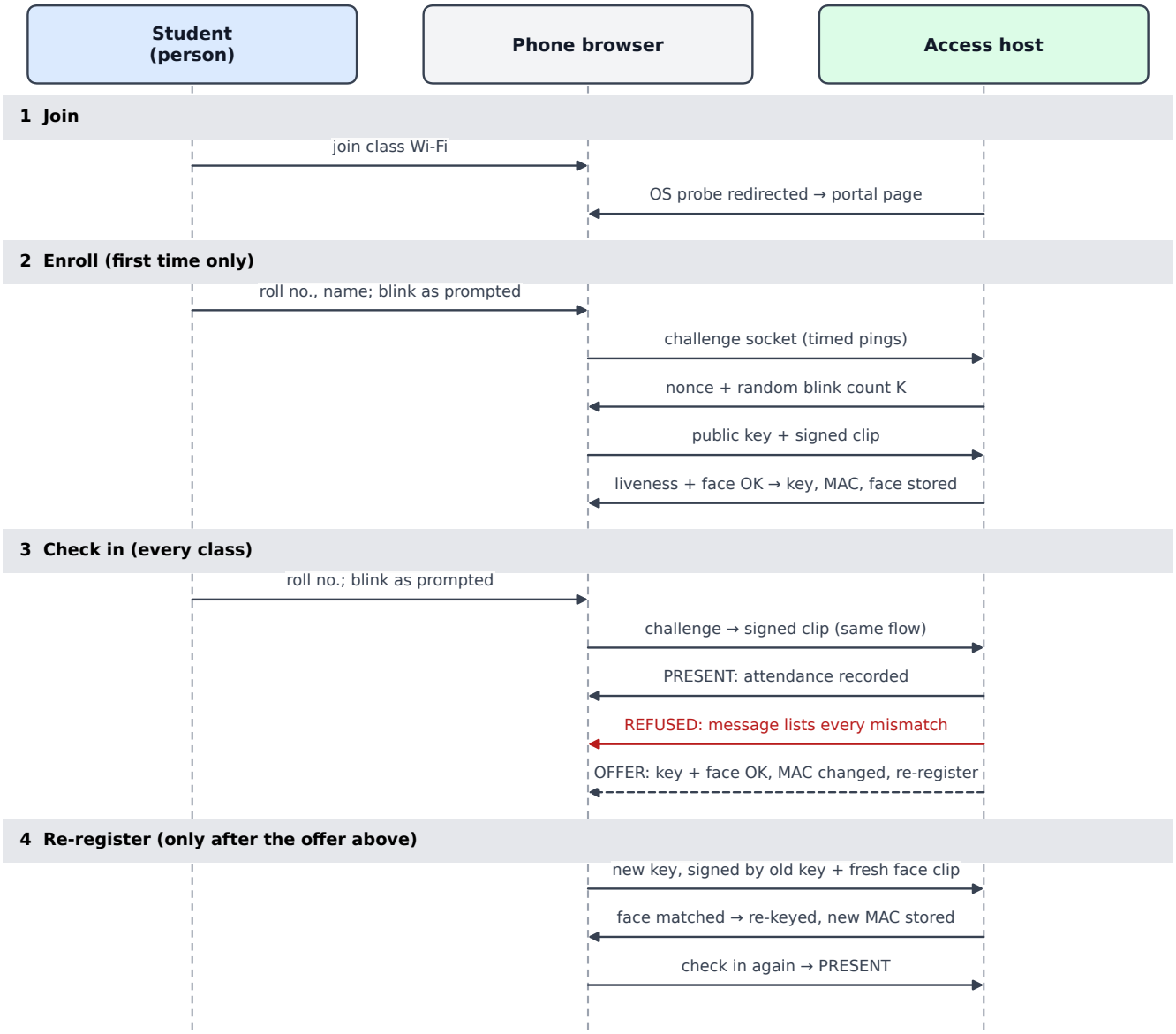
Student (person)
Phone browser
Access host
1 Join
join class Wi-Fi
OS probe redirected → portal page
2 Enroll (first time only)
roll no., name; blink as prompted
challenge socket (timed pings)
nonce + random blink count K
public key + signed clip
liveness + face OK → key, MAC, face stored
3 Check in (every class)
roll no.; blink as prompted
challenge → signed clip (same flow)
PRESENT: attendance recorded
REFUSED: message lists every mismatch
OFFER: key + face OK, MAC changed, re-register
4 Re-register (only after the offer above)
new key, signed by old key + fresh face clip
face matched → re-keyed, new MAC stored
check in again → PRESENT